SIMATS ENGINEERING
ASSESSMENT TOOL 6

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1715**
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COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour
QUESTIONS: 3 **Total Marks:30**

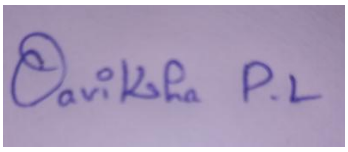
Annexure A

sDry Run Evaluation

Code of Conduct:

I P.L.OAVIKSHA(192572001)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph:

```
      A
     / \
    B   C
   / \ / \
  D  E F  G
```

Task: Starting from node **A**, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration	Queue	Visited Node
1		
2		
3		
4		
5		
6		
7		

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

2. Depth-First Search (DFS) Using the same graph,

```
      A
     / \
    B   C
   / \ / \
  D  E F  G
```

Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

3. **Initial State :**
   1 3 6
   5 0 2
   4 7 8

Goal State :

1	**2**	**3**
4	**5**	**6**
7	**8**	**0**

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.
Fill in the table.

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1			
2			
3			
4			
5			

Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	**20**	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
Criteria	**Weightage (%)**	**Level 4 – Exemplary (4)**	**Level 3 – Proficient (3)**	**Level 2 – Developing (2)**	**Level 1 – Beginning (1)**
State Expansion and Dry Run Execution	**30**	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	**20**	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	**20**	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	**10**	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

Question 1

Breadth-First Search (BFS)

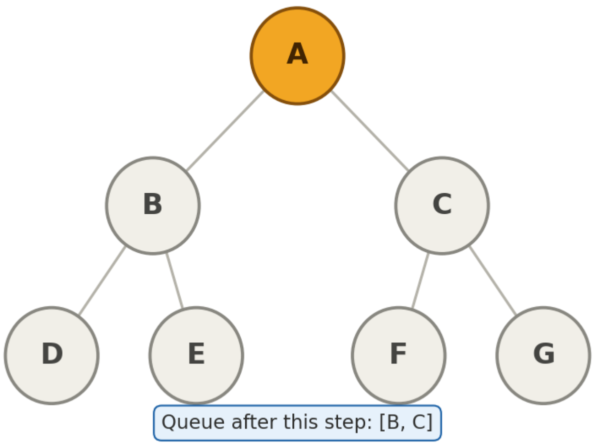
Graph: A connects to B and C. B connects to D and E. C connects to F and G. BFS uses a queue (FIFO): visit a node, then enqueue its unvisited children.

Completed Iteration Table

Iteration	Queue (before expansion)	Visited Node
1	[A]	A
2	[B, C]	B
3	[C, D, E]	C
4	[D, E, F, G]	D
5	[E, F, G]	E
6	[F, G]	F
7	[G]	G

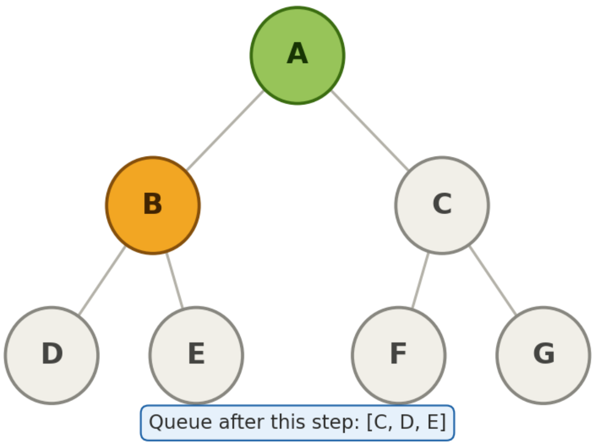
Step-by-Step Diagrams

Step 1: Visit node A



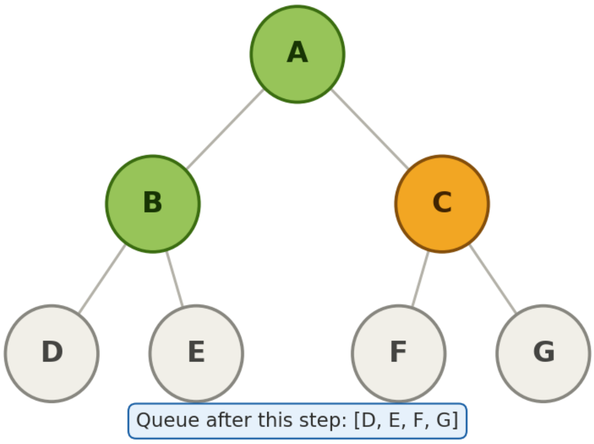
Nodes added to queue: B, C

Step 2: Visit node B



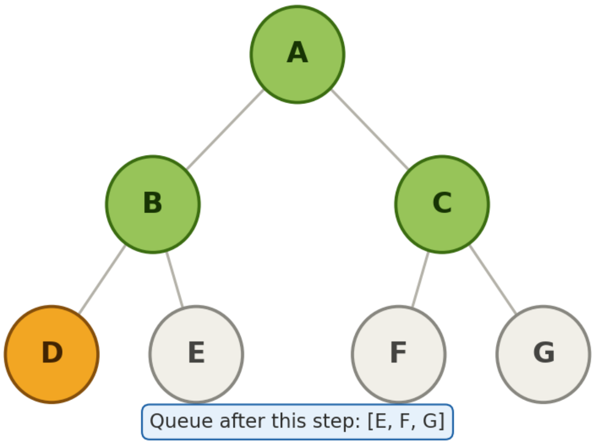
Nodes added to queue: D, E

Step 3: Visit node C



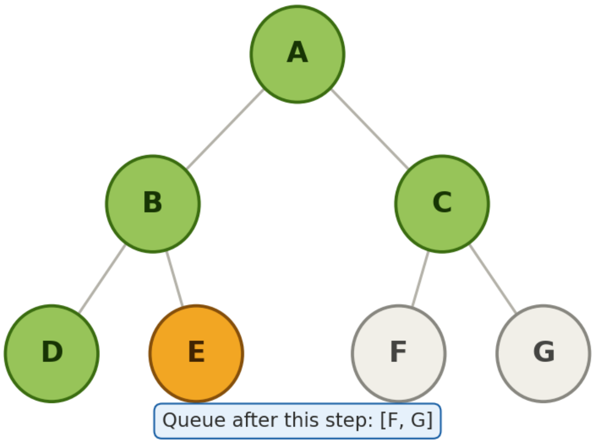
Nodes added to queue: F, G

Step 4: Visit node D



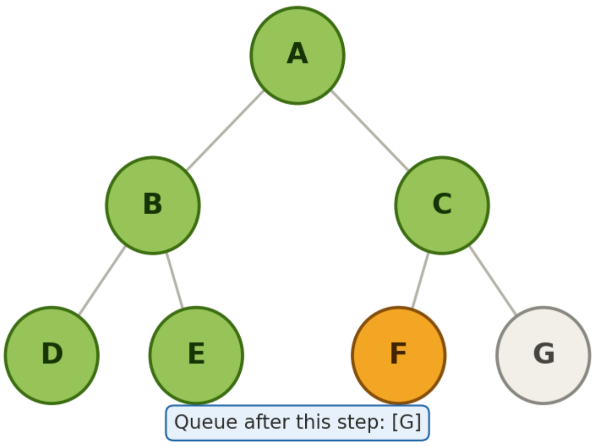
No new nodes added (leaf node).

Step 5: Visit node E



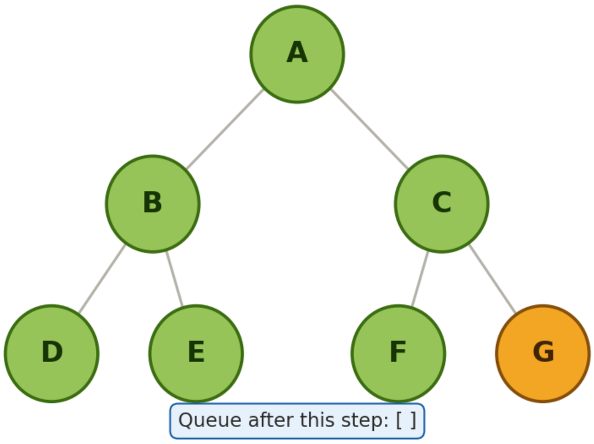
No new nodes added (leaf node).

Step 6: Visit node F



No new nodes added (leaf node).

Step 7: Visit node G



No new nodes added (leaf node).

Answers

1. BFS traversal order: A → B → C → D → E → F → G

2. Queue contents after each iteration: [B,C] → [C,D,E] → [D,E,F,G] → [E,F,G] → [F,G] → [G] → [ ]

Question 2

Depth-First Search (DFS)

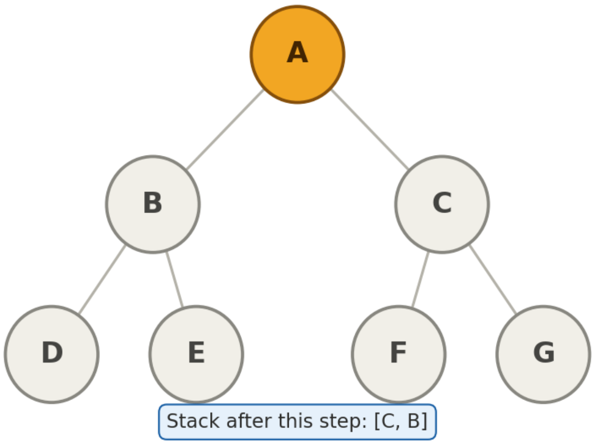
Same graph as Question 1. DFS uses a stack (LIFO). To visit left-to-right, the right child is pushed first, then the left child, so the left child pops off next.

Completed Iteration Table

Iteration	Stack (before expansion)	Visited Node
1	[A]	A
2	[C, B]	B
3	[C, E, D]	D
4	[C, E]	E
5	[C]	C
6	[G, F]	F
7	[G]	G

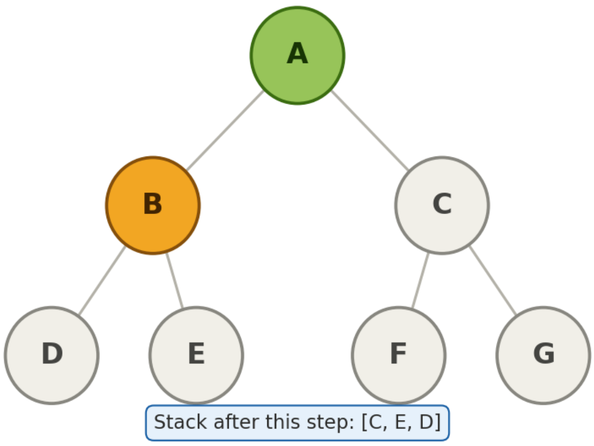
Step-by-Step Diagrams

Step 1: Visit node A



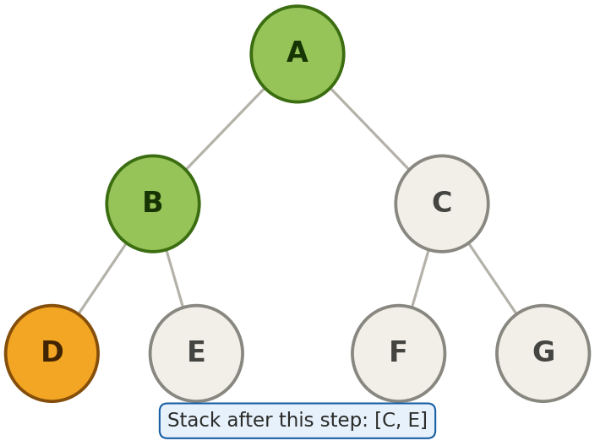
Pushed onto stack: push C, push B

Step 2: Visit node B



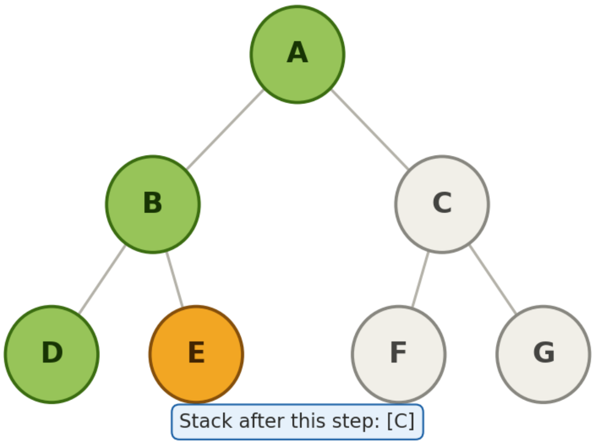
Pushed onto stack: push E, push D

Step 3: Visit node D



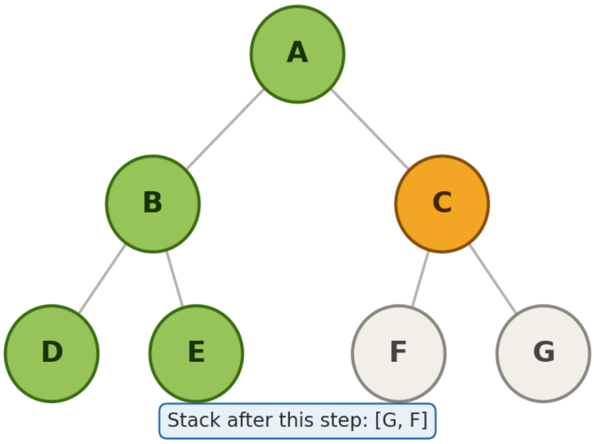
No new nodes pushed (leaf node backtrack next).

Step 4: Visit node E



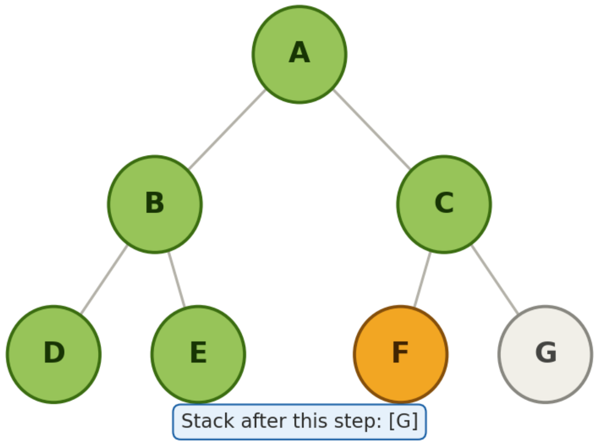
No new nodes pushed (leaf node backtrack next).

Step 5: Visit node C



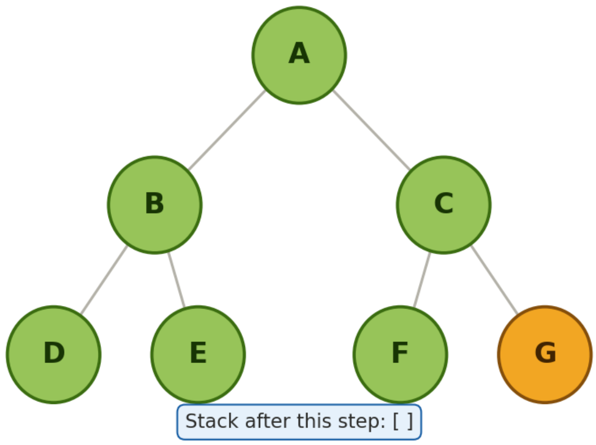
Pushed onto stack: push G, push F

Step 6: Visit node F



No new nodes pushed (leaf node — backtrack next).

Step 7: Visit node G



No new nodes pushed (leaf node — backtrack next).

Answer

DFS traversal order: A → B → D → E → C → F → G

Question 3

8-Puzzle BFS (Initial → Goal)

Initial state: 1 3 6 / 5 0 2 / 4 7 8   Goal state: 1 2 3 / 4 5 6 / 7 8 0   (0 = blank tile)

Verified via an actual BFS solver run on this exact puzzle not estimated. Shortest solution: 8 moves (9 states). Total states generated across the whole search (all branches): 838. Unique states visited before the goal was found: 500. Because BFS expands level-by-level, it must explore every state at depth 1, then depth 2, and so on, before reaching the depth-8 goal — this is why hundreds of states get generated even though the winning path is only 8 moves long

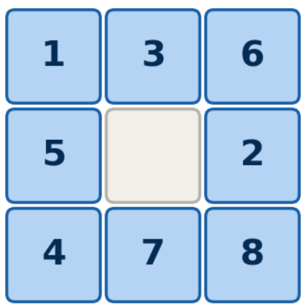
.

Solution-Path Table

Iteration	Current State (before move)	Move Made	Resulting State
1	1 3 6 / 5 0 2 / 4 7 8	Right	1 3 6 / 5 2 0 / 4 7 8
2	1 3 6 / 5 2 0 / 4 7 8	Up	1 3 0 / 5 2 6 / 4 7 8
3	1 3 0 / 5 2 6 / 4 7 8	Left	1 0 3 / 5 2 6 / 4 7 8
4	1 0 3 / 5 2 6 / 4 7 8	Down	1 2 3 / 5 0 6 / 4 7 8
5	1 2 3 / 5 0 6 / 4 7 8	Left	1 2 3 / 0 5 6 / 4 7 8
6	1 2 3 / 0 5 6 / 4 7 8	Down	1 2 3 / 4 5 6 / 0 7 8
7	1 2 3 / 4 5 6 / 0 7 8	Right	1 2 3 / 4 5 6 / 7 0 8
8	1 2 3 / 4 5 6 / 7 0 8	Right	1 2 3 / 4 5 6 / 7 8 0 (GOAL)

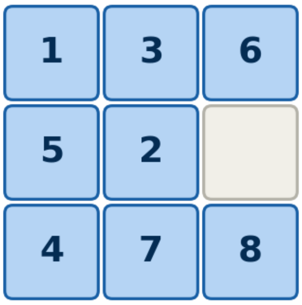
Step-by-Step Diagrams

Step 1: Initial state



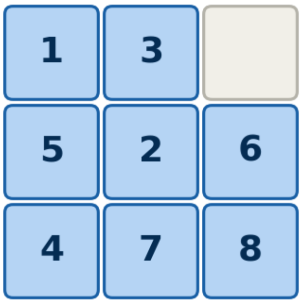
1 3 6 / 5 0 2 / 4 7 8

Step 2: after moving blank Right



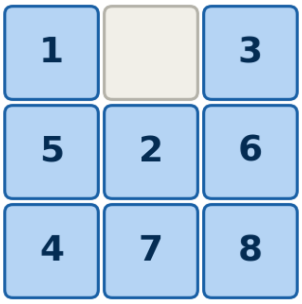
1 3 6 / 5 2 0 / 4 7 8

Step 3: after moving blank Up



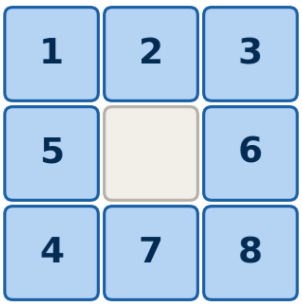
1 3 0 / 5 2 6 / 4 7 8

Step 4: after moving blank Left



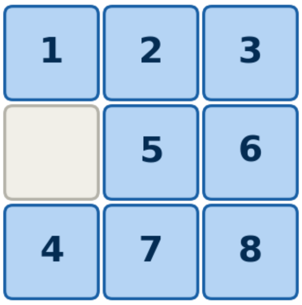
1 0 3 / 5 2 6 / 4 7 8

Step 5: after moving blank Down



1 2 3 / 5 0 6 / 4 7 8

Step 6: after moving blank Left

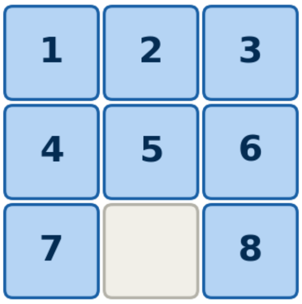


1 2 3 / 0 5 6 / 4 7 8

Step 7: after moving blank Down

1 2 3 / 4 5 6 / 0 7 8

Step 8: after moving blank Right



1 2 3 / 4 5 6 / 7 0 8

Step 9: GOAL STATE reached

1 2 3 / 4 5 6 / 7 8 0 (GOAL)

QUESTIONS:

1. Which node is expanded in each iteration?

In each iteration, the node that gets expanded is the one sitting at the front of the queue. Along the solution path, the states expanded in order are: the initial state, then the state after moving the blank right, then after moving up, then left, then down, then left again, then down again, and finally after moving right, which leads to the goal.

2. How many states are generated in total?

A total of eight hundred and thirty-eight states are generated over the course of the entire search. Out of these, five hundred unique states are actually visited before the goal state is reached.

3. Write the complete solution path from the initial state to the goal state.

The complete solution path starts at the initial state and reaches the goal state in eight moves. The blank tile is moved right, then up, then left, then down, then left again, then down again, then right, and finally right once more, at which point the puzzle reaches the goal configuration.

4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

Breadth-first search always finds the shortest solution in an unweighted eight-puzzle problem because it explores the search space level by level, meaning it fully expands every state at a given depth before moving on to any state at the next depth. Since every move in the puzzle costs exactly the same, the very first time the goal state is reached, it is guaranteed to be at the smallest possible number of moves. No shorter path could still be waiting to be discovered, because breadth-first search would have already found it first.